

3.Design Zero-shot, One-shot, and Few-shot prompts to summarize the following article into 50 words. Compare the generated summaries based on accuracy, completeness, and readability.

ZERO-SHOT:

```
from transformers import pipeline
```

```
generator = pipeline("text-generation", model="gpt2")
```

```
article = """
```

Artificial Intelligence is transforming many industries by improving efficiency, reducing human effort, and enabling faster decision-making. In healthcare, AI helps doctors diagnose diseases, analyze medical images, and recommend treatments.

In education, AI supports personalized learning through intelligent tutoring systems. AI is also widely used in finance for fraud detection and risk analysis.

Despite its benefits, AI raises concerns about privacy, security, and job displacement. Responsible development and ethical use of AI are essential to maximize its benefits while minimizing risks.

```
"""
```

```
prompt = f"""
```

Summarize the following article in exactly 50 words.

Article:

```
{article}
```

Summary:

```
"""
```

```
result = generator(
```

```
    prompt,
```

```
    max_length=180,
```

```
    do_sample=True,
```

```
    temperature=0.5
```

```
)
```

```
print(result[0]["generated_text"])
```

output



The screenshot shows a Jupyter Notebook on the left and a terminal window on the right. The Jupyter Notebook contains the following code:

```
from transformers import pipeline

# Load GPT-2 model
generator = pipeline("text-generation", model="gpt2")

Article = """
Artificial Intelligence is transforming many industries by improving efficiency,
reducing human effort, and enabling faster decision-making. In healthcare, AI
helps doctors diagnose diseases, analyze medical images, and recommend treatments.
In education, AI supports personalized learning through intelligent tutoring
systems. AI is also widely used in finance for fraud detection and risk analysis.
Despite its benefits, AI raises concerns about privacy, security, and job
displacement. Responsible development and ethical use of AI are essential to
maximize its benefits while minimizing risks.
"""

prompt = f"""
Summarize the following article in exactly 50 words.

Article:
{Article}

Summary:
"""

result = generator(
    prompt,
    max_length=160,
    do_sample=True,
    temperature=0.5
)

print(result[0]["generated_text"])
```

The terminal window on the right shows the output of the code, which is a summary of the article in exactly 50 words:

```
Summary:
The rise of artificial intelligence and the rise of AI in business, politics, and
society has a profound impact on the lives of millions of people. The rise of
artificial intelligence and the rise of AI in business, politics, and society ha
s a profound impact on the lives of millions of people. The rise of artificial i
ntelligence and the rise of AI in business, politics, and society has a profound
impact on the lives of millions of people.

Introduction
The world can be divided into two categories:
1. Cyberpunk
2. The Cyberpunk Generation
The Cyberpunk Generation is a group of cyberpunk writers who are working to crea
te a new age of digital society that is both open to all, and free of the oppres
sive state.
In the Cyberpunk Generation, these writers are working to create a new age of di
gital society that is both open to all, and free of the oppressive state.
They are working to create a new age of digital society that is both open to all
, and free of the oppressive state.
The Cyberpunk Generation is a group of cyberpunk writers who are working to crea
te a new age of digital society that is both open to all, and free of the oppres
sive state.
They
>>>
```

ONE-SHOT

from transformers import pipeline

generator = pipeline("text-generation", model="gpt2")

prompt = ""

Example

Article:

Online learning provides students with flexibility and allows them to access educational resources from anywhere. It also reduces travel time and learning costs.

Summary:

Online learning offers flexible, accessible, and cost-effective education by enabling students to learn from anywhere while reducing travel time and expenses.

Now summarize the following article in exactly 50 words.

Article:

Artificial Intelligence is transforming many industries by improving efficiency, reducing human effort, and enabling faster decision-making. In healthcare, AI helps doctors diagnose diseases, analyze medical images, and recommend treatments. In education, AI supports personalized learning through intelligent tutoring systems. AI is also used in finance for fraud detection and risk analysis. Despite its benefits, AI raises concerns about privacy, security, and job displacement. Responsible development and ethical use of AI are essential.

Summary:

""

result = generator(

prompt,

```

max_length=250,

do_sample=True,

temperature=0.5

)

print(result[0]["generated_text"])

output:

```

The screenshot shows a Jupyter Notebook on the left and an IDLE Shell window on the right. The Jupyter Notebook contains the following code:

```

from transformers import pipeline

generator = pipeline("text-generation", model="gpt2")

prompt = """
Example

Article:
Online learning provides students with flexibility and allows them to access educational resources from anywhere. It also reduces travel time and learning costs.

Summary:
Online learning offers flexible, accessible, and cost-effective education by enabling students to learn from anywhere while reducing travel time and expenses.

Now summarize the following article in exactly 50 words.

Article:
Artificial Intelligence is transforming many industries by improving efficiency, reducing human effort, and enabling faster decision-making. In healthcare, AI helps doctors diagnose diseases, analyze medical images, and recommend treatments. In education, AI supports personalized learning through intelligent tutoring systems. AI is also used in finance for fraud detection and risk analysis. Despite its benefits, AI raises concerns about privacy, security, and job displacement. Responsible development and ethical use of AI are essential.

Summary:
"""

result = generator(
    prompt,
    max_length=250,
    do_sample=True,
    temperature=0.5
)

print(result[0]["generated_text"])

```

The IDLE Shell window on the right displays the output of the code, showing the generated text for the prompt. The output includes the example article and summary, followed by the generated summary for the second article.

FEW-SHOT:

```

from transformers import pipeline

generator = pipeline("text-generation", model="gpt2")

prompt = """

```

Example 1

Article:

Online learning provides students with flexibility and allows them to access educational resources from anywhere.

Summary:

Online learning offers flexible and accessible education for students.

Example 2

Article:

Renewable energy sources such as solar and wind power reduce pollution and help protect the environment.

Summary:

Renewable energy reduces pollution and supports environmental sustainability.

Now summarize the following article in exactly 50 words.

Article:

Artificial Intelligence is transforming many industries by improving efficiency, reducing human effort, and enabling faster decision-making. In healthcare, AI helps doctors diagnose diseases, analyze medical images, and recommend treatments. In education, AI supports personalized learning through intelligent tutoring systems. AI is also used in finance for fraud detection and risk analysis. Despite its benefits, AI raises concerns about privacy, security, and job displacement. Responsible development and ethical use of AI are essential.

Summary:

"""

result = generator(

prompt,

max_length=300,

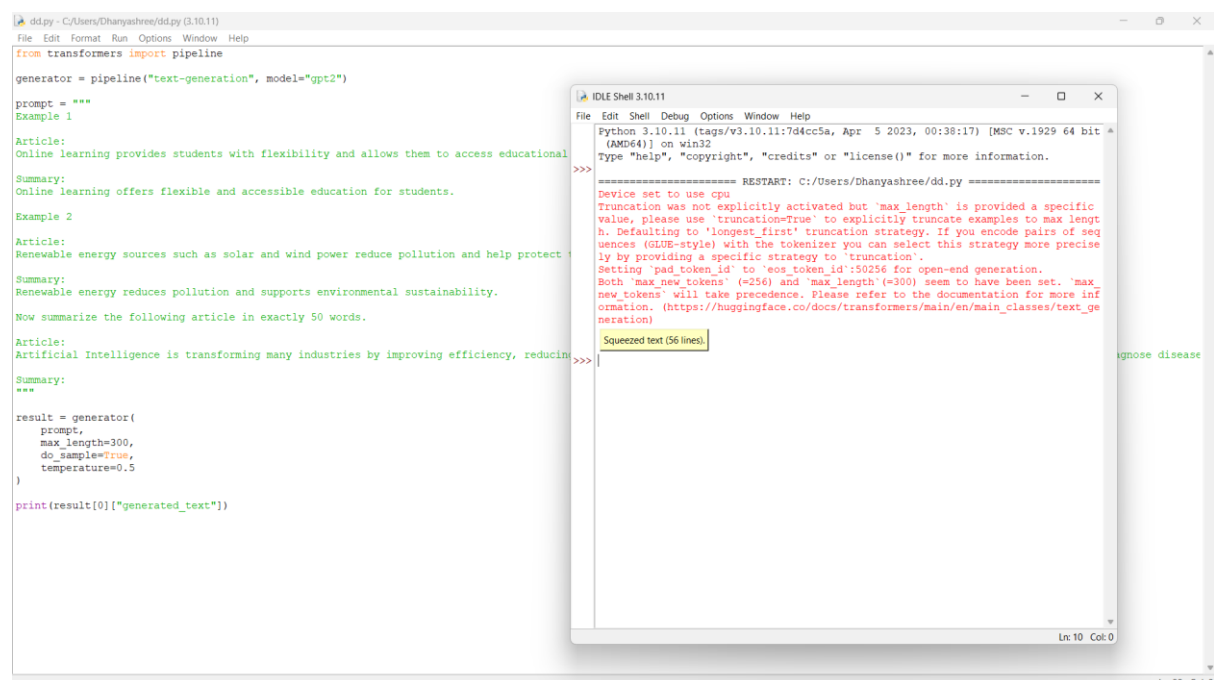
do_sample=True,

temperature=0.5

)

print(result[0]["generated_text"])

output:



The image shows a Python script in an IDE window titled 'dd.py - C:/Users/Dhanyashree/dd.py (3.10.11)'. The script imports the 'pipeline' function from 'transformers', creates a generator for 'text-generation' using the 'gpt2' model, and defines a prompt. It then calls the generator with the prompt, max_length=300, do_sample=True, and temperature=0.5, and prints the generated text. The output is shown in a terminal window titled 'IDLE Shell 3.10.11'. The terminal output shows the prompt, the generated text, and a 'Squeezed text (56 lines)' message.

```
dd.py - C:/Users/Dhanyashree/dd.py (3.10.11)
File Edit Format Run Options Window Help
from transformers import pipeline

generator = pipeline("text-generation", model="gpt2")

prompt = """
Example 1
Article:
Online learning provides students with flexibility and allows them to access educational resources.
Summary:
Online learning offers flexible and accessible education for students.
Example 2
Article:
Renewable energy sources such as solar and wind power reduce pollution and help protect the environment.
Summary:
Renewable energy reduces pollution and supports environmental sustainability.
Now summarize the following article in exactly 50 words.
Article:
Artificial Intelligence is transforming many industries by improving efficiency, reducing human effort, and enabling faster decision-making. In healthcare, AI helps doctors diagnose diseases, analyze medical images, and recommend treatments. In education, AI supports personalized learning through intelligent tutoring systems. AI is also used in finance for fraud detection and risk analysis. Despite its benefits, AI raises concerns about privacy, security, and job displacement. Responsible development and ethical use of AI are essential.
Summary:
"""

result = generator(
    prompt,
    max_length=300,
    do_sample=True,
    temperature=0.5
)

print(result[0]["generated_text"])
```

```
IDLE Shell 3.10.11
File Edit Shell Debug Options Window Help
Python 3.10.11 (tags/v3.10.11:7d4cc5a, Apr 5 2023, 00:38:17) [MSC v.1929 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: C:/Users/Dhanyashree/dd.py =====
Device set to use cpu
Truncation was not explicitly activated but 'max_length' is provided a specific value, please use 'truncation=True' to explicitly truncate examples to max length. Defaulting to 'longest_first' truncation strategy. If you encode pairs of sequences (GLUE-style) with the tokenizer you can select this strategy more precisely by providing a specific strategy to 'truncation'.
Setting 'pad_token_id' to 'eos_token_id':50256 for open-end generation.
Both 'max_new_tokens' (=256) and 'max_length' (=300) seem to have been set. 'max_new_tokens' will take precedence. Please refer to the documentation for more information. (https://huggingface.co/docs/transformers/main/en/main_classes/text_generation)
Squeezed text (56 lines):
>>>
```

